Theo Di Piazza

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Personal page, **LinkedIn**

PhD machine learning researcher specializing in computer vision and NLP, with 6 first-author publications and industry research experience with Centre Leon Berard, Thales, Hospices Civil de Lyon and Philips.

Education

- **PhD in Computed Science, University of Lyon** — Lyon, France
  Doctor of Philosophy in Computer Science — *01/2024 – 01/2027*
  Research areas: Natural Language Processing, Computer Vision, Multimodal Learning, 3D Medical Imaging
- **MSc in Applied Mathematics, Ecole Normale Superieure Paris-Saclay** — Paris, France
  Master of Science in Mathematics and Computer Science (MVA); Highest Honors — *09/2022 – 09/2023*
  Relevant courses: Optimization, Machine Learning, Deep Learning, Time Series, Graphs, Computer Vision
- **MEng in Applied Mathematics, INSA Rennes** — Rennes, France
  Engineering Degree in Applied Mathematics — *09/2017 – 09/2022*
  Relevant courses: Machine Learning, Linear Algebra, Stochastic Models, Statistical Learning, Computer Science

Professional Experience

- **Hospices Civil de Lyon** — Lyon, France
  Machine Learning Research Scientist - PhD in collaboration with Philips Clinical Informatics — *01/2024 – 01/2027*
  - **Outcome**: Developed deep learning architectures and analysis tools to study representation learning in 3D computed tomography scans with applications to abnormality classification and automated report generation, including language models, vision-language models and foundation models. Led to 5 first author publications.
  - **Skills**: codebase development, large-scale experiments, precise ablations, autonomy, communication.
  - **Advisors**: Prof. Loic Boussel, Carole Lazarus, Olivier Nempont.
- **Thales** — Elancourt, France
  Machine Learning Research Intern — *04/2023 – 09/2023*
  - **Outcome**: Developed deep learning and computer-vision methods for UAV localization, using a contour-based feature extraction approach that improved localization performance. Led to a first-author publication.
  - **Advisors**: Prof. Gabriele Facciolo, Prof. Enric Leinhardt-Llopis, Corentin Abgral, Benedicte Bascle.
- **Centre Leon Berard** — Lyon, France
  Applied Machine Learning Intern — *03/2022 – 08/2022*
  - **Outcome**: Evaluated oncology survival prediction models under cross-institutional distribution shifts.
  - **Advisor**: Loic Verlingue.
- **Hupi** — Bidart, France
  Data Scientist Intern — *05/2021 – 09/2021*
  - **Outcome**: Developed and deployed a machine failure prediction model for a Spanish industrial group, and a predictive model for forecasting institutionalization dates among dependent individuals.

Selected Publications

- **2026 - MICCAI Spotlight**: UniCT: A Unified Joint Multi-Task Framework for 3D Chest CT Abnormality Analysis.
- **2026 - MICCAI**: ChronoSurv: A Clinical Pathway-Guided Graph Framework for Multimodal Survival Analysis.
- **2026 - MELBA**: Structured Spectral Graph Representation Learning for Abnormality Analysis from 3D CT Scans.
- **2025 - MIDL Oral**: A Global-Local Attention Model for 3D CT Volumes Abnormality Classification.
- **2025 - ISBI Oral**: Automated Abnormality-Guided Report Generation from 3D Chest CT Volumes.
- **2024 - IGARSS Oral**: Leveraging Edge Detection and Neural Networks for better UAV Localization.

Skills

- **Programming**: Python, SQL, C++, R
- **Frameworks**: PyTorch, NumPy, Scikit-Learn, HuggingFace Transformers
- **Applications**: Computer Vision, Natural Language Processing (NLP), Large Language Models, Vision-Language Models, Multimodal Learning, Cross-Modal Retrieval, Image Captioning, Pre-training, Fine-tuning, Data Visualization
- **Tools**: Git, Linux, LaTeX, Adobe Photoshop & After Effects
- **Language**: French (native), English (professional proficiency), Spanish (elementary proficiency)